Laxmi Charitable Trust's

Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce

Department of Information Technology (B.Sc.I.T SemesterV)

Applied Business

Practical – X

Roll No.:T019	Name: Krishna Mishra
Class: TYIT	Batch:1
Date of Assignment:26-09-26	Date/Time of Submission:26-09-26

Practical 10: Regression Analysis and Time Series Analysis Part A:

Regression Analysis

1. Load the sales_data.csv dataset.

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import r2_score, mean_squared_error
```

14.7s

```
# 1. Load dataset
df = pd.read_csv("sales_data.csv")
```

0.0s

2. Display the first five records.

```
# 2. Display first five records
print("First Five Records:")
print(df.head())

# 3 & 4. Independent and dependent variables
X = df[["advertising"]]
y = df["sales"]

print("\nIndependent Variable: Advertising")
print("Dependent Variable: Sales")
```

```
First Five Records:
   advertising  sales
0           10     15
1           12     18
2           15     20
3           18     24
4           20     27

Independent Variable: Advertising
Dependent Variable: Sales
```

3. Identify the independent and dependent variables.

```
# 3 & 4. Independent and dependent variables
X = df[["advertising"]]
y = df["sales"]
```

```
Independent Variable: Advertising
Dependent Variable: Sales
```

4. Select advertising as the independent variable and sales as the dependent variable.

```
Independent Variable: Advertising
Dependent Variable: Sales
```

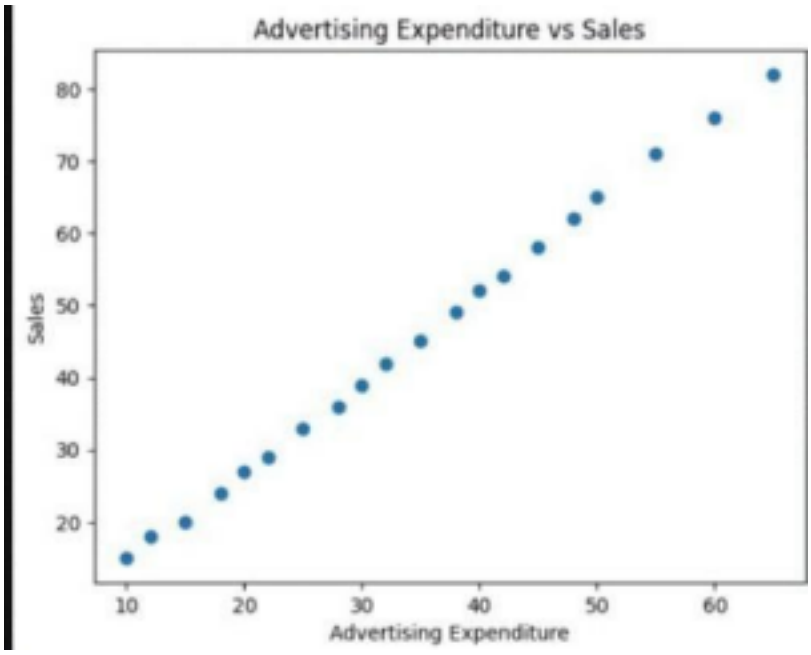
5. Check the data for missing values.

```
# 5. Check missing values
print("\nMissing Values:")
print(df.isnull().sum())
```

```
Missing Values:
advertising    0
sales          0
dtype: int64
```

6. Create a scatter plot of advertising expenditure and sales.

```
# 6. Scatter plot
plt.scatter(df["advertising"], df["sales"])
plt.xlabel("Advertising Expenditure")
plt.ylabel("Sales")
plt.title("Advertising Expenditure vs Sales")
plt.show()
✓ 0.2s
```



7. Develop a simple linear regression model.

```
# 7. Develop Simple Linear Regression Model
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

# 8. Regression equation
b0 = model.intercept_
b1 = model.coef_[0]

print("\nRegression Equation:")
print(f"Sales = {b0:.2f} + {b1:.2f}(Advertising)")
```

0.4s

```
Regression Equation:
Sales = 1.75 + 1.24(Advertising)
```

8. Display the regression equation.

```
# 8. Regression equation
b0 = model.intercept_
b1 = model.coef_[0]

print("\nRegression Equation:")
print(f"Sales = {b0:.2f} + {b1:.2f}(Advertising)")
```

0.4s

```
Sales = 1.75 + 1.24(Advertising)
```

9. Use the model to predict sales for a given advertising expenditure.

```
# 9. Predict sales for a given advertising expenditure
advertising_value = 50
predicted_sales = model.predict([[advertising_value]])

print(f"\nPredicted Sales for Advertising = {advertising_value}: "
      f"{predicted_sales[0]:.2f}")

# Predictions for test data
y_pred = model.predict(X_test)
```

```
Predicted Sales for Advertising = 50: 63.96
```

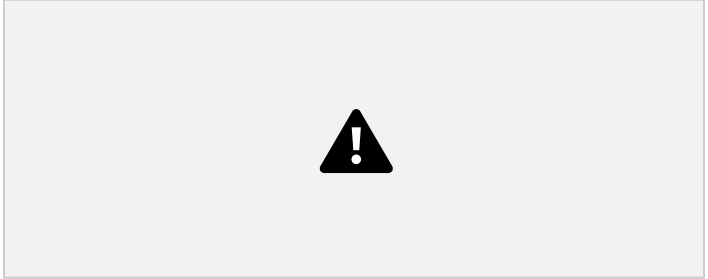
10. Calculate the R^2 score of the model.



11. Calculate the Mean Squared Error (MSE).

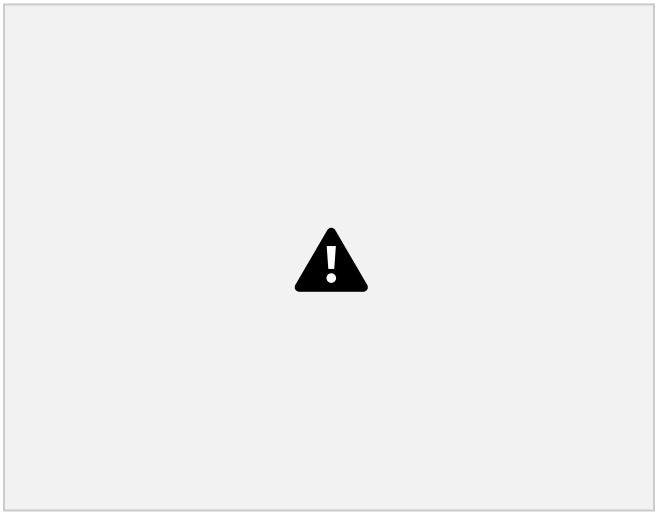


12. Compare the actual sales with the predicted sales

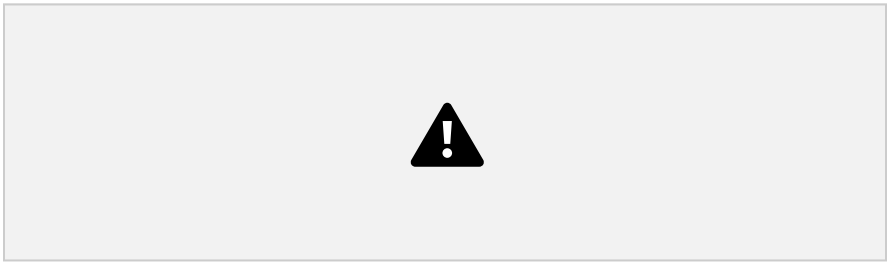


13. Plot the regression line with the actual data points.



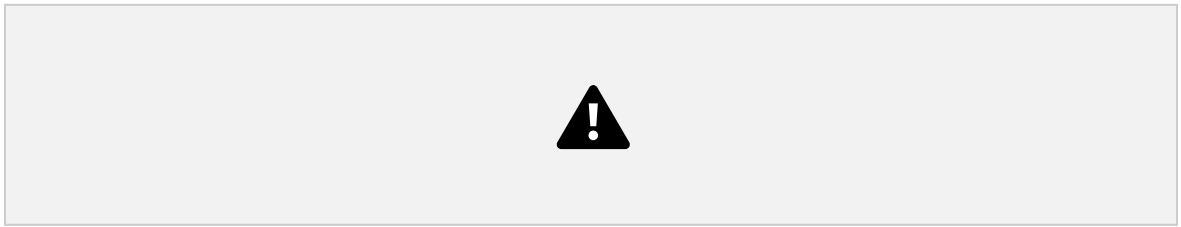


14. Interpret the regression results.





15. Write a conclusion about how advertising expenditure is related to sales. A simple regression model is:





$= 0 +$

$1_{()}$

:

16. Load monthly sales data.

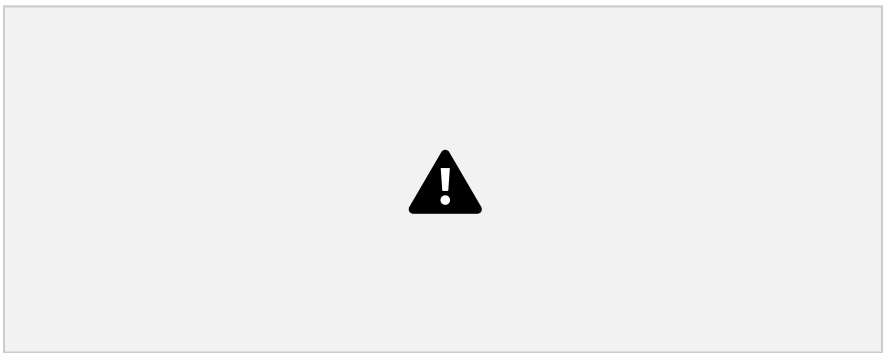


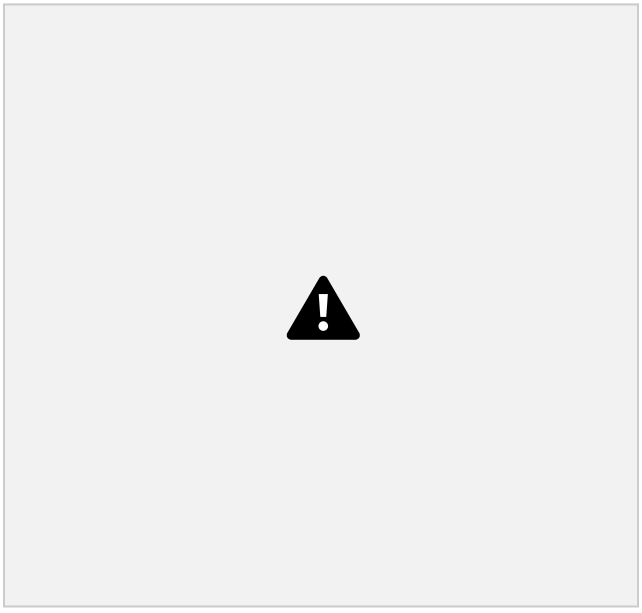
17. Convert the date column into date format.



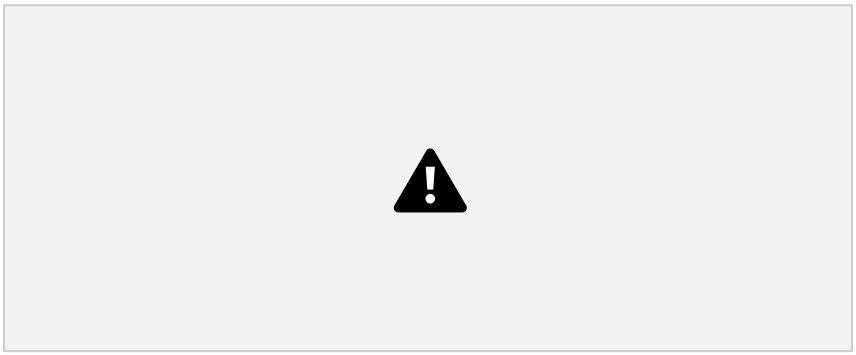


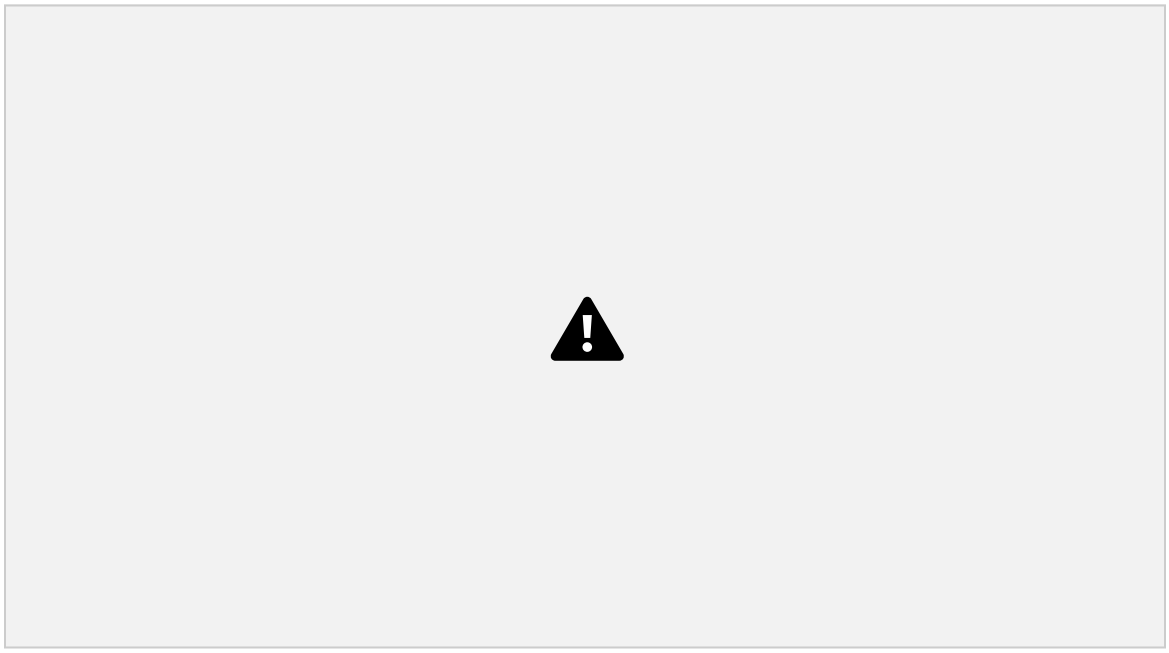
18. Plot monthly sales using a line chart.



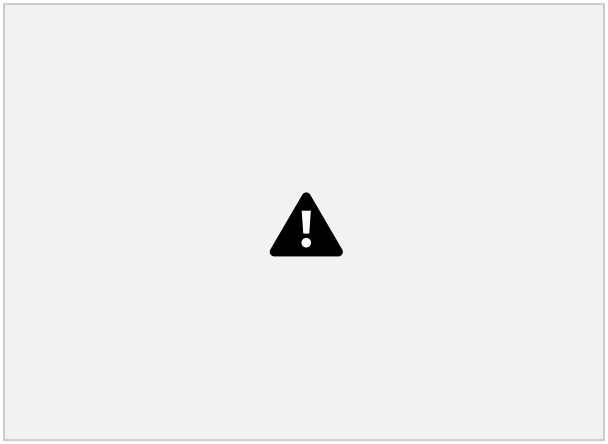


19. Observe the trend.





20. Apply Exponential Smoothing.

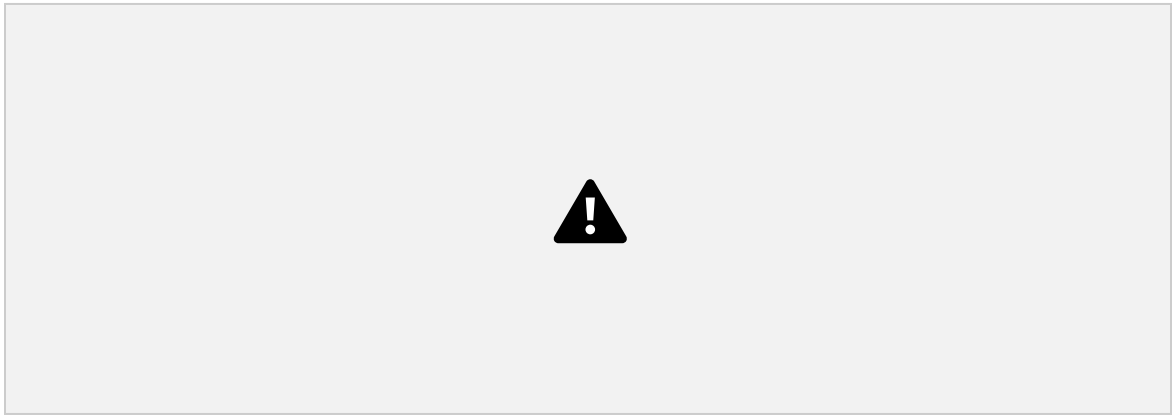


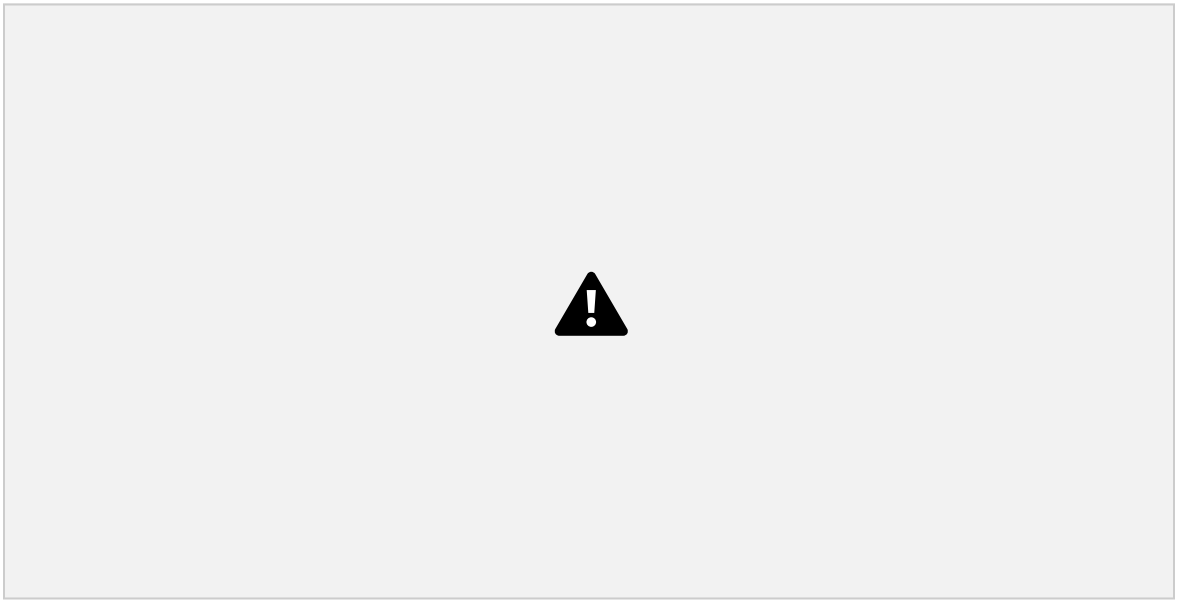


21. Forecast the next 3 months.

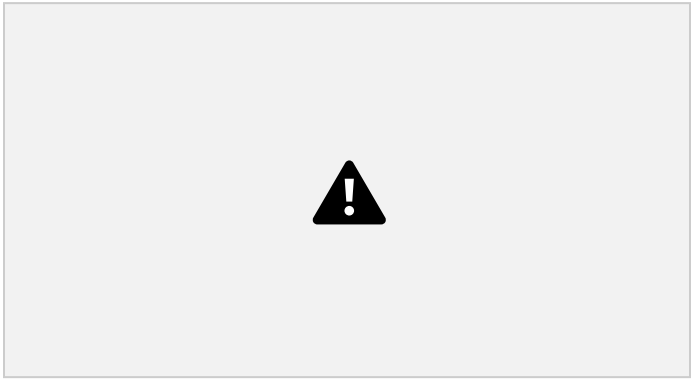


22. Plot actual and forecasted sales.



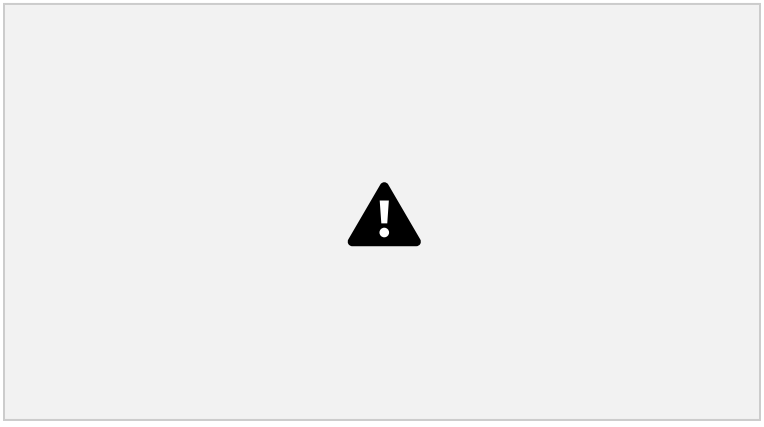


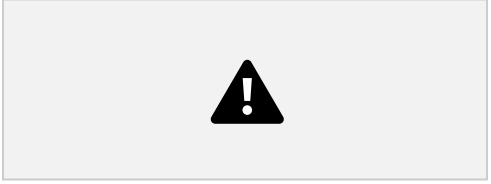
23. Calculate forecasting error.





24. Apply ARIMA to the same data.





25. Identify which model gives the lower forecasting error for the given dataset. 29. Write the final conclusion.

